

Harshita Karnam

Full-Stack Developer | Reinforcement Learning Researcher

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LeetCode: leetcode.com/u/HarshitaKA | Portfolio: portfolio.vercel.app

PROFESSIONAL SUMMARY

Co-author on SafeGAT, an LLM-augmented Graph Attention RL framework for adaptive traffic signal control that cut average travel time by 36% versus fixed-time control, currently under review at Expert Systems with Applications. Third-year B.Tech CSE student who has shipped five full-stack and ML projects end-to-end, from system design through production deployment (Next.js, React, Node.js, MongoDB), paired with production engineering experience at Atomprix Technologies.

EDUCATION

B.Tech — Computer Science & Engineering

Indian Institute of Information Technology, Sri City, India

2024 – 2028

CGPA: 8.5/10

RESEARCH

SafeGAT (*Under Review, Expert Systems with Applications — Co-Author*) — Uncertainty-gated, safety-constrained LLM supervision for Graph Attention RL in adaptive traffic signal control. Reduced average travel time by 36% vs. fixed-time control, outperforming GAT-DQN/CoLight baselines on SUMO 4×4/7×28 networks; cut unnecessary LLM interventions ~4× while eliminating safety violations; validated on real-world CityFlow datasets (Hangzhou, Jinan, New York).

TECHNICAL SKILLS

- **Languages:** Python, C, C++, HTML, CSS, JavaScript, TypeScript
- **Web / Full-Stack:** React, Next.js, Node.js, REST APIs, JWT, MongoDB, Firebase, Vercel
- **ML / RL:** PyTorch, OpenAI Gym, SUMO, DQN, DDQN, PPO, CoLight, Multi-Agent RL
- **Concepts:** Data Structures & Algorithms, Graph Attention Networks, Prompt Engineering
- **Tools:** Git / GitHub, Google Cloud Platform (GCS), Linux, VS Code

EXPERIENCE

Full Stack Developer Intern — Atomprix Technologies Pvt Ltd (Vahni AI)

Jun 2026 – Jul 2026

- Redesigned the LLM prompt powering `generateProjectMetadata` to fetch a user's existing project names by type and constrain generation against duplicates.
- Shipped end-to-end voice input for the chatbox across dashboard and app studio (mic capture, browser speech-to-text streaming into chat) plus an attachment icon with GCS-backed file uploads.
- Implemented backend logic to auto-start a user's cloud workstation (sandbox) on app open from the dashboard, checking run state and provisioning on demand instead of requiring a fresh workstation per session.

LEADERSHIP & ACTIVITIES

Director of Management, E-Cell — IIIT Sri City: Marketed three entrepreneurship events (SellFiesta, Catalyst, Arthashastra), driving attendance of ~100 students at each

Chief Editor, Insiiits Magazine — IIIT Sri City: Led a team of 7 students on Issue 7, growing readership to 400 students across campus

Member, Developer Relations, GDG — Co-organized TechSprint and Chakravyuh

PROJECTS

Velora Market — Multi-Vendor Marketplace

TypeScript, Next.js, MongoDB, JWT

- Architected a full-stack marketplace with distinct buyer, seller, and admin dashboards, role-based authentication, and a Next.js API backend using HTTP-only refresh-token cookies for secure session handling.
- [GitHub: harshitak13/Velora](https://github.com/harshitak13/Velora) [Live: velora-three-irid.vercel.app](https://velora-three-irid.vercel.app)

CinePlex — React Movie Discovery App

React, Firebase, TMDB API

- Built a trending-film browser with genre search, trailers, a personal watchlist, infinite scroll, and actor pages, backed by Firebase Auth and Firestore for persistent user data.
- [GitHub: harshitak13/cineplex](https://github.com/harshitak13/cineplex) [Live: cineplex-aupw.vercel.app](https://cineplex-aupw.vercel.app)

SafeGAT — LLM-Integrated RL Traffic Signal Control

PyTorch, SUMO, LLM

- Designed a hybrid LLM + Graph Attention RL architecture for adaptive traffic control, benchmarked against DQN, DDQN, PPO, and CoLight on single and 4×4 grids, improving generalization to unseen intersections via LLM-derived policy guidance.
- [GitHub: harshitak13/SafeGat](https://github.com/harshitak13/SafeGat)